

ENGR&204 — VISUAL GUIDE: match the picture, copy the template

How to use this on the exam (3 steps): 1. **Look** at your exam circuit → find the picture below that matches it. 2. **Copy** that template line-by-line, filling the blanks with your numbers. 3. **Check** your answer's shape against the curve picture.

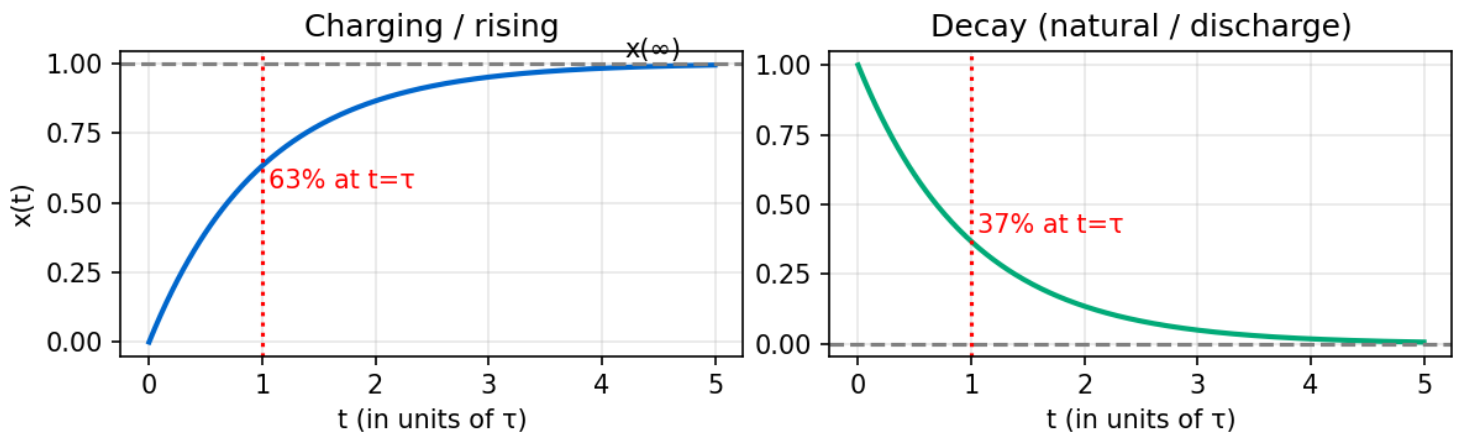
First, count the energy-storage elements (capacitors + inductors):

0 caps/inductors → DC problem	→ use TYPE 5/6 (nodal, mesh, Thévenin, op-amp)
1 cap OR 1 inductor → FIRST-ORDER	→ use TYPE 1 (RC) or TYPE 2 (RL) ← most of your exam
1 cap AND 1 inductor → SECOND-ORDER	→ use TYPE 3 (series RLC) or TYPE 4 (parallel RLC)

THE ONE FORMULA you use for every first-order problem

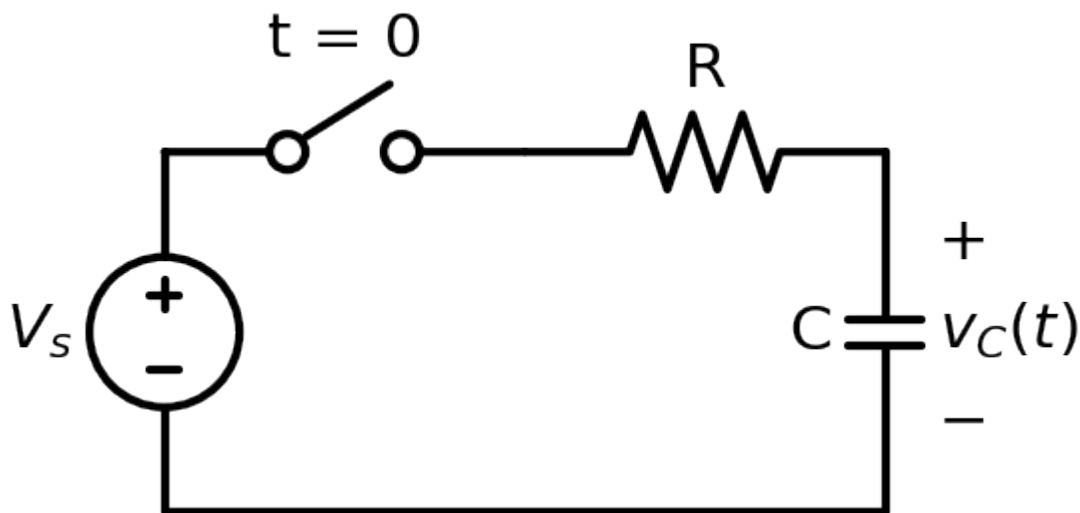
$$x(t) = \underbrace{x(\infty)}_{\text{final}} + \left[\underbrace{x(0^+)}_{\text{initial}} - \underbrace{x(\infty)}_{\text{final}} \right] \cdot e^{\left(- \underbrace{t}_{\text{time constant}} / \tau \right)}$$

x = the $v(t)$ or $i(t)$ you're asked for. You only ever need **3 numbers**: $x(0^+)$, $x(\infty)$, τ .



Left = answer rises to a final value (charging). Right = answer decays to 0 (natural/discharge). After 1τ you've moved 63% of the way; after 5τ it's basically done.

TYPE 1 — FIRST-ORDER RC (one capacitor)



Recognize it: a **capacitor**, a switch, resistors, a DC source. Asked for $v_C(t)$ or $v(t)$.

COPY THIS TEMPLATE (fill the blanks):

STEP 1 $t < 0$ (before switch): capacitor = OPEN. Solve DC for the cap voltage.
 $v_C(0^-) = \underline{\hspace{2cm}} \rightarrow v_C(0^+) = v_C(0^-) = \underline{\hspace{2cm}}$ (cap voltage can't jump)

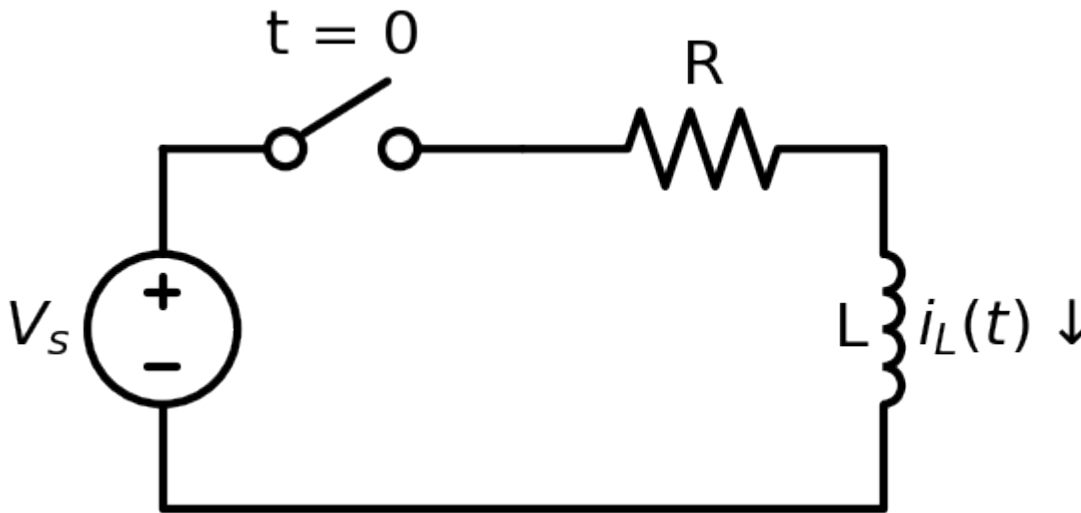
STEP 2 $t \rightarrow \infty$ (after switch settles): capacitor = OPEN. Solve DC again.
 $v_C(\infty) = \underline{\hspace{2cm}}$

STEP 3 Time constant: turn OFF independent sources (V-source $\rightarrow$ wire, I-source $\rightarrow$ gap).
 Look into the capacitor's two terminals, find the resistance: $R_{th} = \underline{\hspace{2cm}}$
 $\tau = R_{th} \cdot C = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ s}$

STEP 4 Plug in:
 $v_C(t) = v_C(\infty) + [v_C(0^+) - v_C(\infty)] e^{(-t/\tau)}$
 $v_C(t) = \underline{\hspace{2cm}} + (\underline{\hspace{2cm}} - \underline{\hspace{2cm}}) e^{(-t/ \underline{\hspace{2cm}})} \text{ V}$

Worked (your HW6 #2): $v_C(0^+) = 4 \text{ V}$, $v_C(\infty) = 6 \text{ V} \Rightarrow v(t) = 6 + (4-6)e^{(-t/\tau)} = 6 - 2e^{(-t/\tau)} \text{ V}$.

TYPE 2 — FIRST-ORDER RL (one inductor)



Recognize it: an **inductor**, a switch, resistors, a DC source. Asked for $i_L(t)$ or $i(t)$.

COPY THIS TEMPLATE:

STEP 1 $t < 0$: inductor = SHORT (plain wire). Solve DC for the inductor current.
 $i_L(0^-) = \underline{\hspace{2cm}} \rightarrow i_L(0^+) = i_L(0^-) = \underline{\hspace{2cm}}$ (inductor current can't jump)

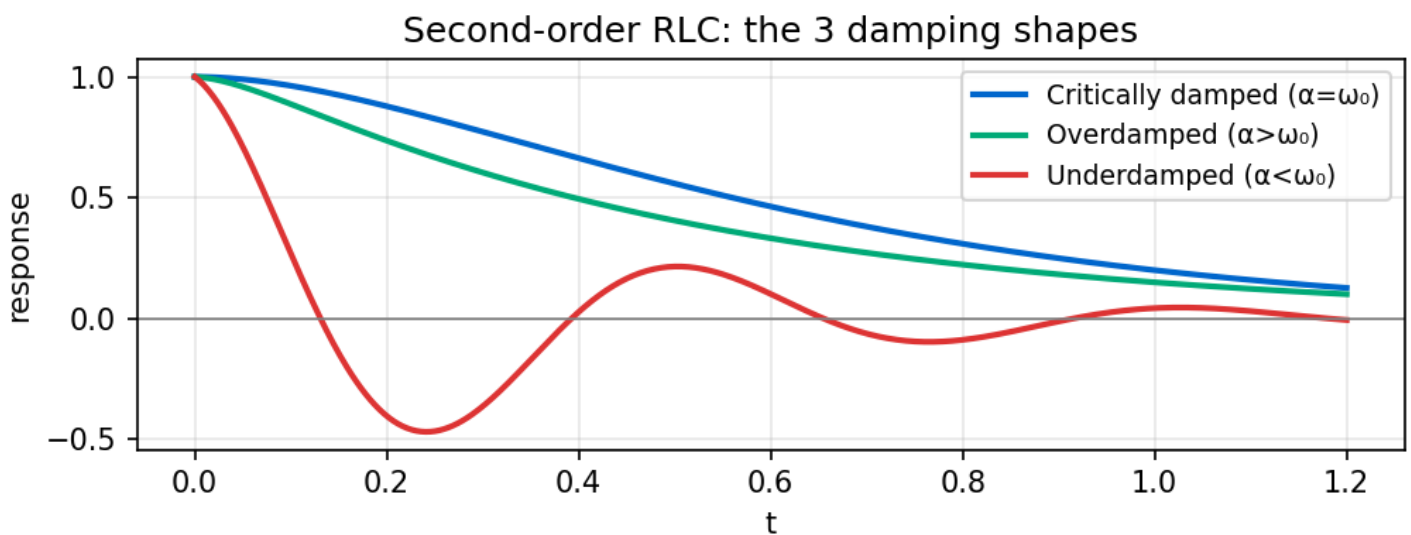
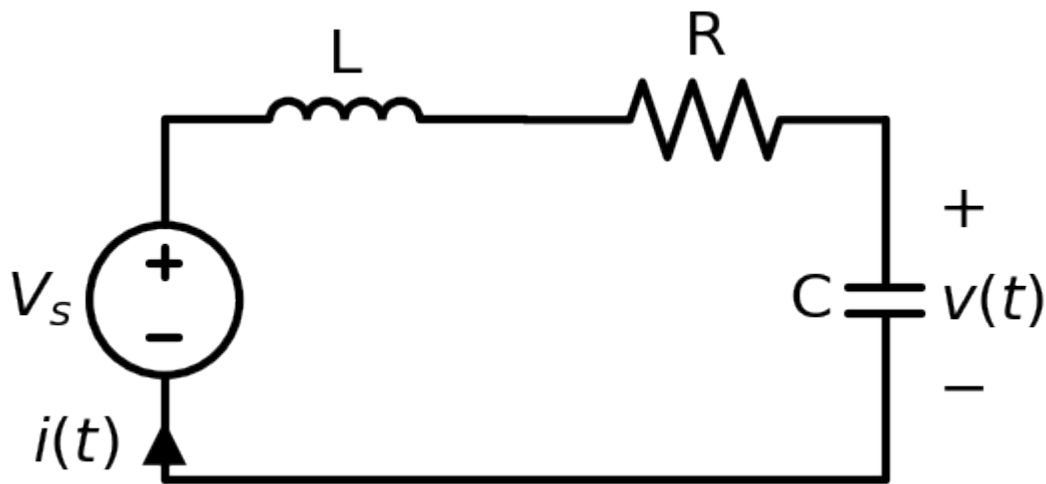
STEP 2 $t \rightarrow \infty$: inductor = SHORT. Solve DC again.
 $i_L(\infty) = \underline{\hspace{2cm}}$

STEP 3 Time constant: turn OFF independent sources. Resistance seen by the inductor:
 $R_{th} = \underline{\hspace{2cm}} \quad \tau = L / R_{th} = \underline{\hspace{2cm}} / \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ s}$

STEP 4 Plug in:
 $i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)] e^{(-t/\tau)}$
 $i_L(t) = \underline{\hspace{2cm}} + (\underline{\hspace{2cm}} - \underline{\hspace{2cm}}) e^{(-t/ \underline{\hspace{2cm}})} \text{ A}$

Need a voltage or another current? Get it from $v_L = L \cdot di_L/dt$, or from Ohm's/KVL/KCL using $i_L(t)$. **Worked (Dorf P8.3-25):** read off the plot $v(0^+) = 100$, $v(\infty) = 20$, point $(0.14\text{s}, 60) \Rightarrow \tau = 0.2\text{s}$, $v(t) = 20 + 80e^{(-5t)} \text{ V}$.

TYPE 3 — SECOND-ORDER SERIES RLC (R, L, C in one loop)



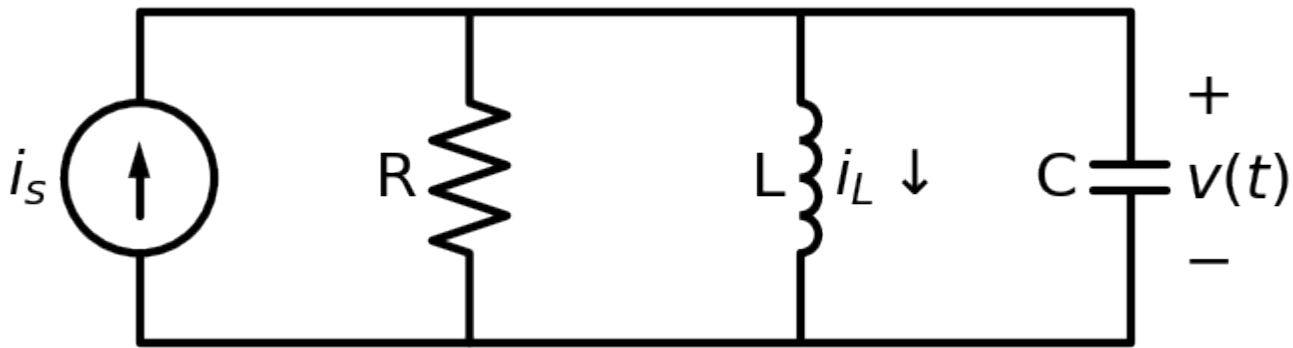
Recognize it: both an inductor and a capacitor **in a single loop** (after turning off sources). Usually solve for loop current $i(t)$.

COPY THIS TEMPLATE:

STEP 1 $\alpha = R / (2L) = \underline{\hspace{2cm}}$ (SERIES: R over 2L)
 STEP 2 $\omega_0 = 1/\sqrt{LC} = \underline{\hspace{2cm}}$
 STEP 3 Characteristic eqn: $s^2 + (R/L)s + 1/(LC) = 0 \rightarrow s^2 + \underline{\hspace{1cm}} s + \underline{\hspace{1cm}} = 0$
 Roots: $s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = \underline{\hspace{2cm}}, \underline{\hspace{2cm}}$
 STEP 4 Pick the case by comparing α vs ω_0 :
 $\alpha > \omega_0$ OVERDAMPED $\rightarrow x(t) = A_1 e^{(s_1 t)} + A_2 e^{(s_2 t)}$
 $\alpha = \omega_0$ CRITICAL $\rightarrow x(t) = (A_1 + A_2 t) e^{(-\alpha t)}$
 $\alpha < \omega_0$ UNDERDAMPED $\rightarrow x(t) = e^{(-\alpha t)} (B_1 \cos \omega_d t + B_2 \sin \omega_d t), \omega_d = \sqrt{(\omega_0^2 - \alpha^2)} = \underline{\hspace{2cm}}$
 STEP 5 Add forced part $x(\infty)$ if a source stays on; find the 2 constants from:
 $x(0^+) = \underline{\hspace{2cm}}$ and $x'(0^+) = \underline{\hspace{2cm}}$ ($x'(0) = i_C(0)/C$ or $v_L(0)/L$)

Worked (handout P1): $R=40, L=0.1, C=1/3 \text{ mF} \Rightarrow \alpha=200, \omega_0=173 \Rightarrow s^2+400s+30000=0$, roots **-100, -300** (overdamped).

TYPE 4 — SECOND-ORDER PARALLEL RLC (R, L, C all in parallel)



Recognize it: inductor and capacitor **side-by-side (parallel)** after turning off sources. Usually solve for node voltage $v(t)$.

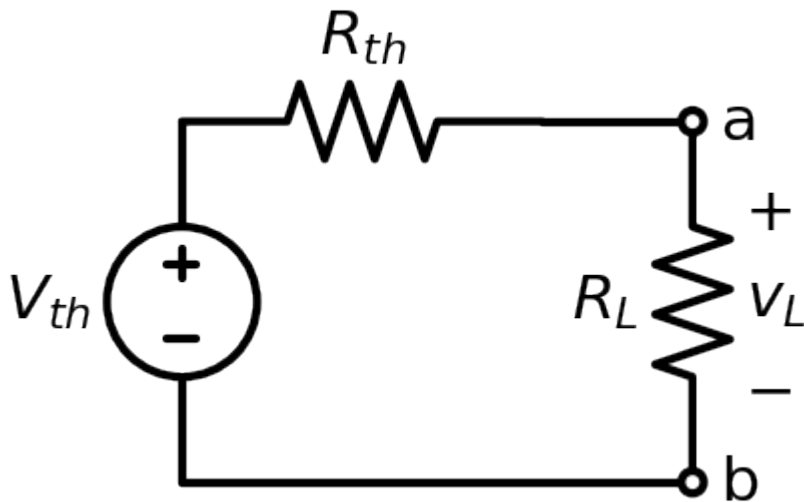
COPY THIS TEMPLATE (only Step 1 changes vs series!):

STEP 1 $\alpha = 1 / (2RC) = \underline{\hspace{2cm}}$ (PARALLEL: 1 over 2RC)
 STEP 2 $\omega_0 = 1/\sqrt{LC} = \underline{\hspace{2cm}}$ (same as series)
 STEP 3 Characteristic eqn: $s^2 + (1/RC)s + 1/(LC) = 0 \rightarrow s^2 + \underline{\hspace{2cm}} s + \underline{\hspace{2cm}} = 0$
 Roots: $s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = \underline{\hspace{2cm}}, \underline{\hspace{2cm}}$
 STEP 4-5 same damping cases & constants as TYPE 3.

Worked (handout P3): $R=10, L=1/2, C=5 \text{ mF} \Rightarrow \alpha=10, \omega_0=20 \Rightarrow s^2+20s+400=0$, roots $-10 \pm j17.3$ (underdamped).

DAMPING TRAP: in **series**, a large R is overdamped; in **parallel**, a small R is overdamped. (Series: $R > 2\sqrt{L/C}$ over; Parallel: $R < 2\sqrt{L/C}$ over.)

TYPE 5 — DC NETWORK: Thévenin / Norton / Max Power

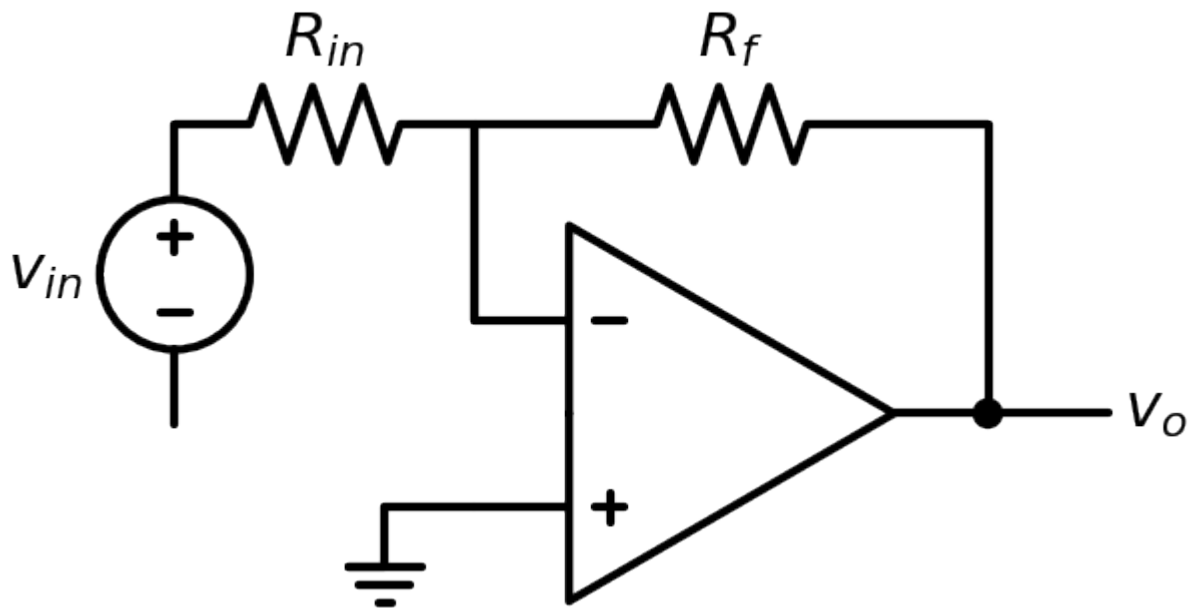


Recognize it: "find the equivalent seen by R_L ", "max power", or a big resistor network. **Replace everything left of a-b with V_{th} in series with R_{th} .**

COPY THIS TEMPLATE:

V_{th} = open-circuit voltage across a-b (remove R_L , find V_{oc}) = $\underline{\hspace{2cm}}$
 R_{th} = (turn off independent sources) resistance looking into a-b = $\underline{\hspace{2cm}}$
 [if dependent sources: $R_{th} = V_{oc} / I_{sc}$]
 Current in a load R: $i = V_{th} / (R_{th} + R) = \underline{\hspace{2cm}}$
 MAX POWER: happens when $R_L = R_{th}$; $P_{max} = V_{th}^2 / (4 R_{th}) = \underline{\hspace{2cm}}$

TYPE 6 — IDEAL OP-AMP



Recognize it: a triangle symbol. **Two golden rules:** (1) no current into the inputs $i_+ = i_- = 0$; (2) $v_+ = v_-$ (virtual short).

COPY THIS TEMPLATE:

Find v_- ($= v_+$, often $= 0$ "virtual ground" if $+$ is grounded).

Write KCL at the inverting ($-$) node using $i_+ = i_- = 0$:

$$(v_{in} - v_-)/R_{in} = (v_- - v_o)/R_f$$

Solve for v_o .

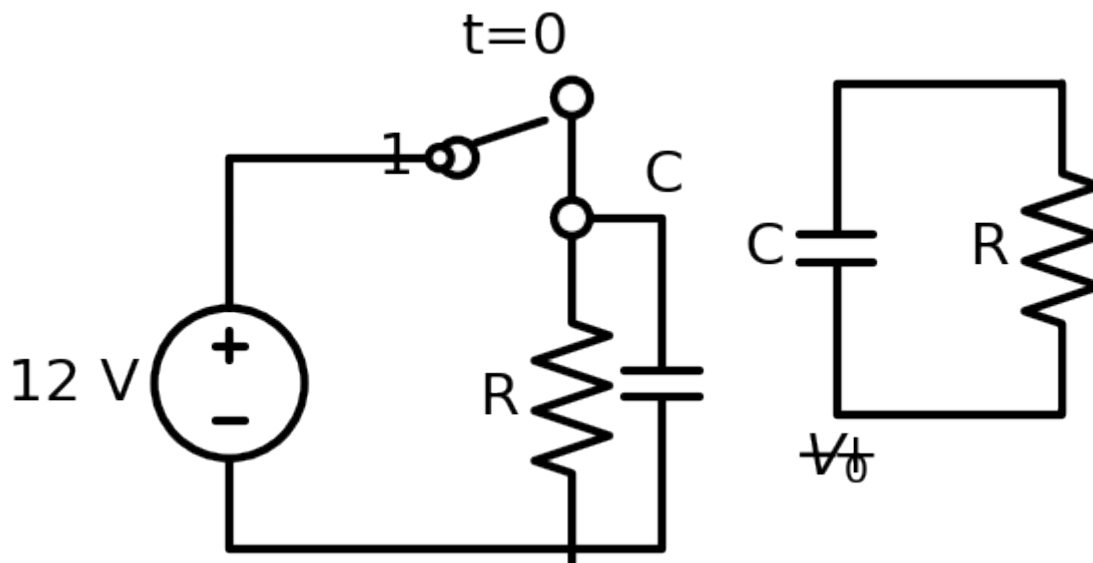
Common results: Inverting $v_o = -(R_f/R_{in}) v_{in}$

Non-invert $v_o = (1 + R_f/R_{in}) v_{in}$

Output current i_o : KCL at the output node.

Worked (your HW2/HW5): $20\text{k}\Omega$ in, $20\text{k}\Omega \rightarrow V_M = -4\text{ V}$; $-2 - v_o = 8 \cdot 3.5 \rightarrow v_o = -30\text{ V}$.

TYPE 7 — TWO-POSITION SWITCH / SOURCE-FREE (natural) transient



Recognize it: switch moves **1**→**2** at $t=0$, OR a charged C / energized L is left alone with just R (no source). This is the same first-order machine — the only change is $x(\infty)$.

COPY THIS TEMPLATE:

Position 1 "for a long time" → find the stored state at the instant of switching:

$v_C(0) = \underline{\hspace{2cm}}$ (cap charged through pos-1 path, DC) / $i_L(0) = \underline{\hspace{2cm}}$

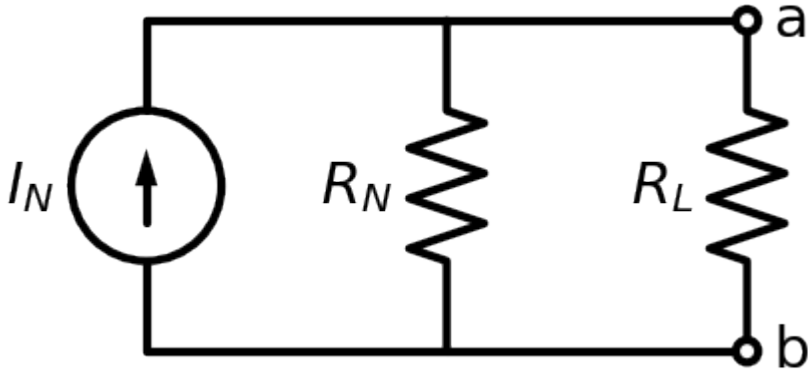
After switch to position 2 (or source removed):

If NO source remains: $x(\infty) = 0 \rightarrow x(t) = x(0) e^{(-t/\tau)}$ (pure decay)

$\tau = R_{th} \cdot C$ or L/R_{th} using the position-2 resistances = $\underline{\hspace{2cm}}$

Energy released: $W_C = \frac{1}{2} C v_C(0)^2$ or $W_L = \frac{1}{2} L i_L(0)^2 = \underline{\hspace{2cm}} \text{ J}$

TYPE 8 — NORTON & SOURCE TRANSFORMATION



Recognize it: "Norton equivalent", or you want to simplify many sources/resistors.

COPY THIS TEMPLATE:

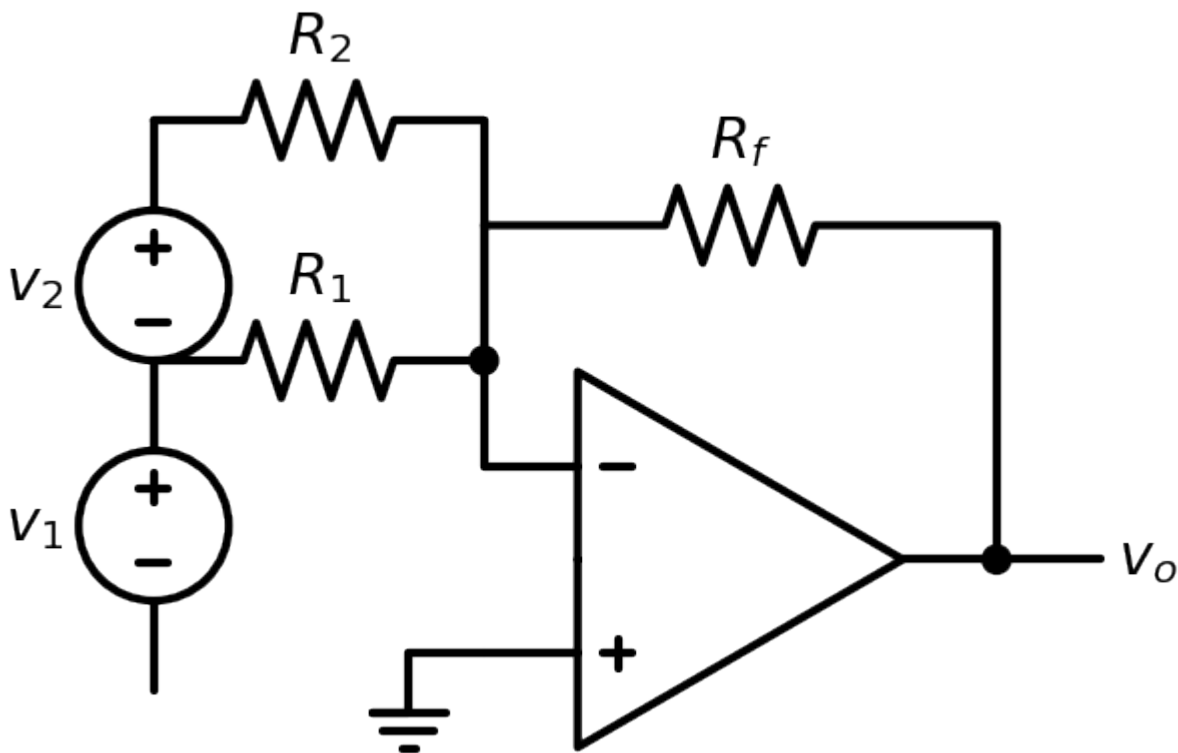
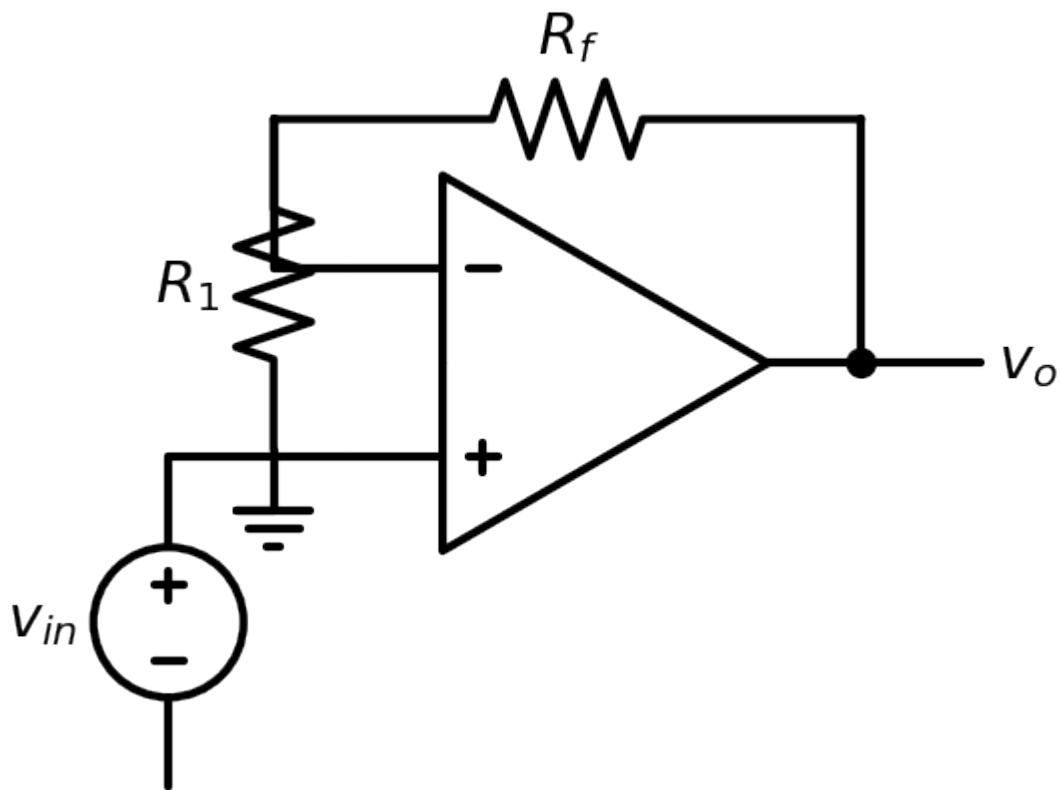
I_N = short-circuit current across a-b (put a wire on a-b, find i) = $\underline{\hspace{2cm}}$

$R_N = R_{th}$ = (independent sources off) resistance into a-b = $\underline{\hspace{2cm}}$

Norton $\ncong$ Thévenin: $V_{th} = I_N \cdot R_{th}$, $I_N = V_{th} / R_{th}$, $R_N = R_{th}$

Source transform: $[V_s \text{ in series with } R] \ncong [I_s = V_s/R \text{ in parallel with } R]$

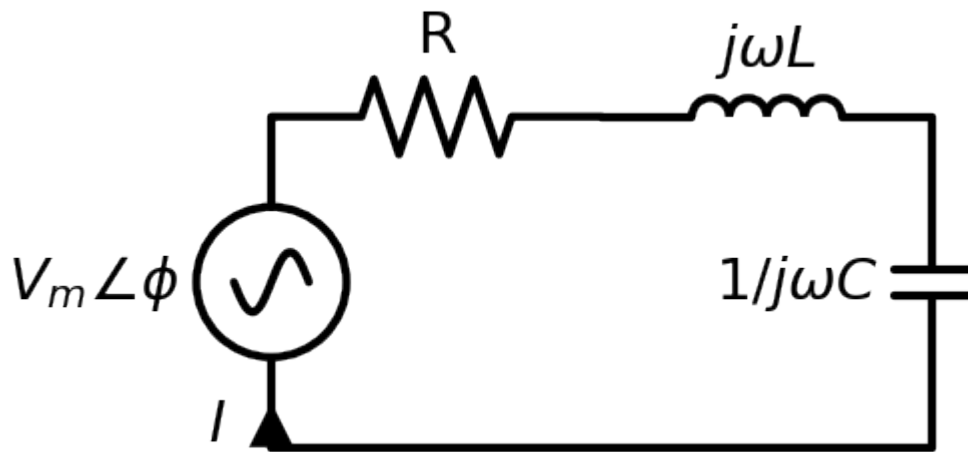
TYPE 9 — OP-AMP VARIANTS



Golden rules: $i_+ = i_- = 0$ and $v_+ = v_-$. Then write KCL at the $-$ node.

Inverting: $v_o = -(R_f/R_{in}) v_{in}$
 Non-inverting: $v_o = (1 + R_f/R_{in}) v_{in}$
 Buffer: $v_o = v_{in}$
 Summing: $v_o = -R_f (v_1/R_1 + v_2/R_2 + \dots)$
 Difference: $v_o = (R_f/R_1)(v_2 - v_1)$ [matched ratios]
 Cascade: multiply the stage gains
 Output limited by supply rails: $V^- \leq v_o \leq V^+$ (saturation)

TYPE 10 — AC STEADY-STATE (phasors)



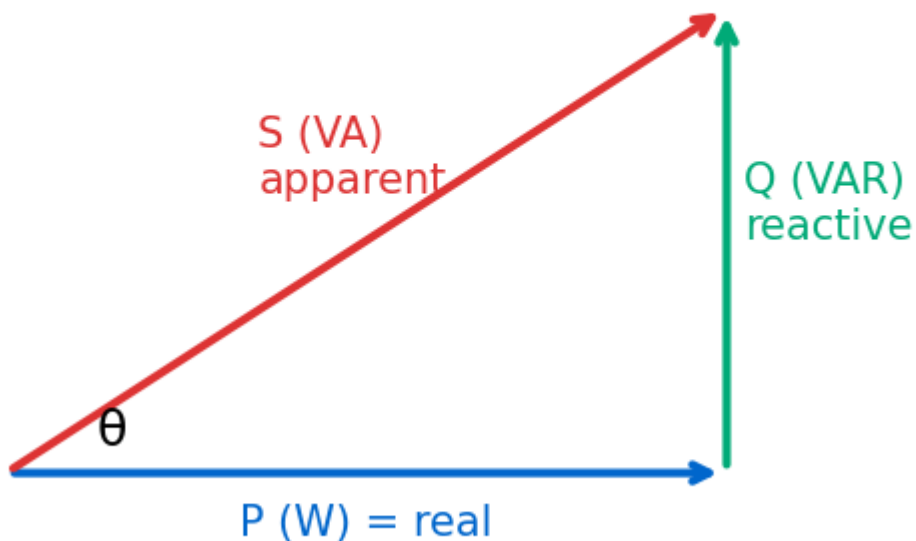
Recognize it: a sinusoidal source $V_m \cos(\omega t + \phi)$; asked for steady-state $v(t)/i(t)$.

COPY THIS TEMPLATE:

1. Source $\rightarrow$ phasor: $V_m \cos(\omega t + \phi) \Rightarrow V = V_m \angle \phi$
 2. Elements $\rightarrow$ impedance at this ω : $Z_R = R$, $Z_L = j\omega L$, $Z_C = 1/(j\omega C) = -j/(\omega C)$
 3. Solve like DC but COMPLEX (Ohm $V = IZ$, dividers, nodal/mesh):
 $I = V / Z_{\text{total}} = \underline{\hspace{1cm}} \angle \underline{\hspace{1cm}}$
 4. Back to time domain: $I_m \angle \theta \Rightarrow i(t) = I_m \cos(\omega t + \theta)$
- Complex math: add/sub in $a + jb$; mult/div in $r \angle \theta$ (mags $\times/\div$, angles $+/-$)

TYPE 11 — AC POWER & POWER-FACTOR CORRECTION

Power triangle: $S = P + jQ$, $\text{pf} = \cos\theta = P/S$

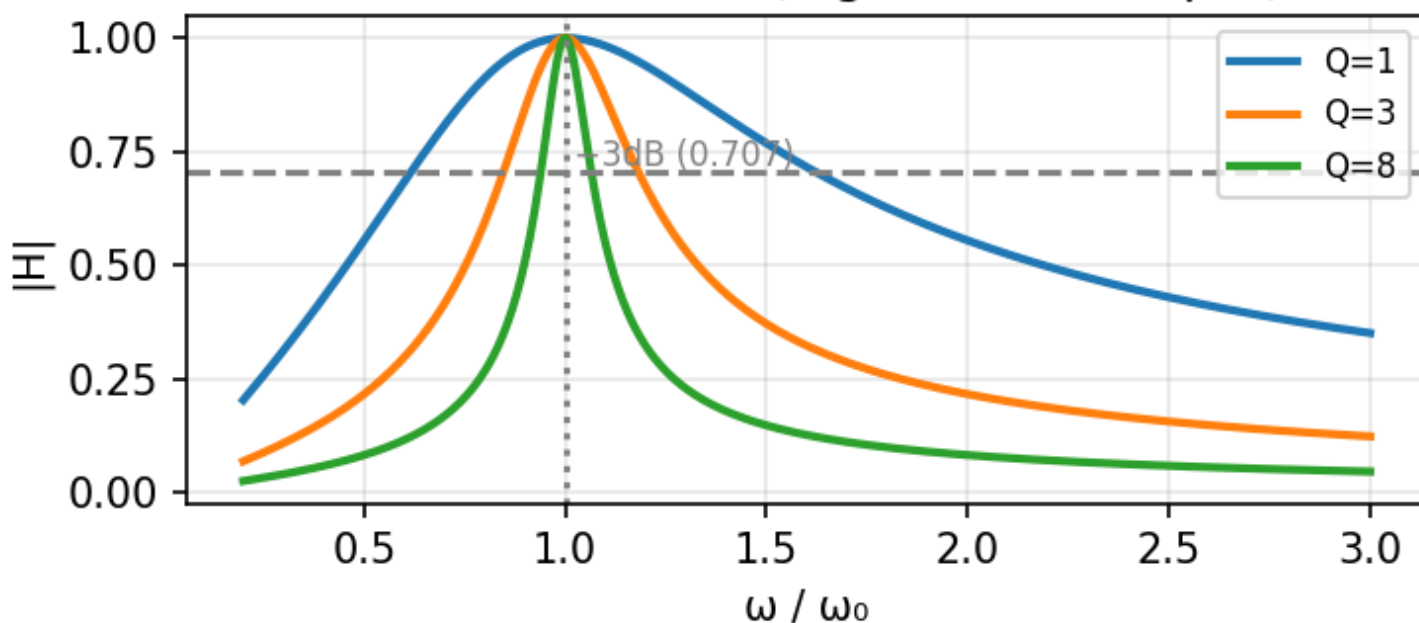


Use RMS values ($V_{\text{rms}} = V_m/\sqrt{2}$). With $\theta = \theta_v - \theta_i$:

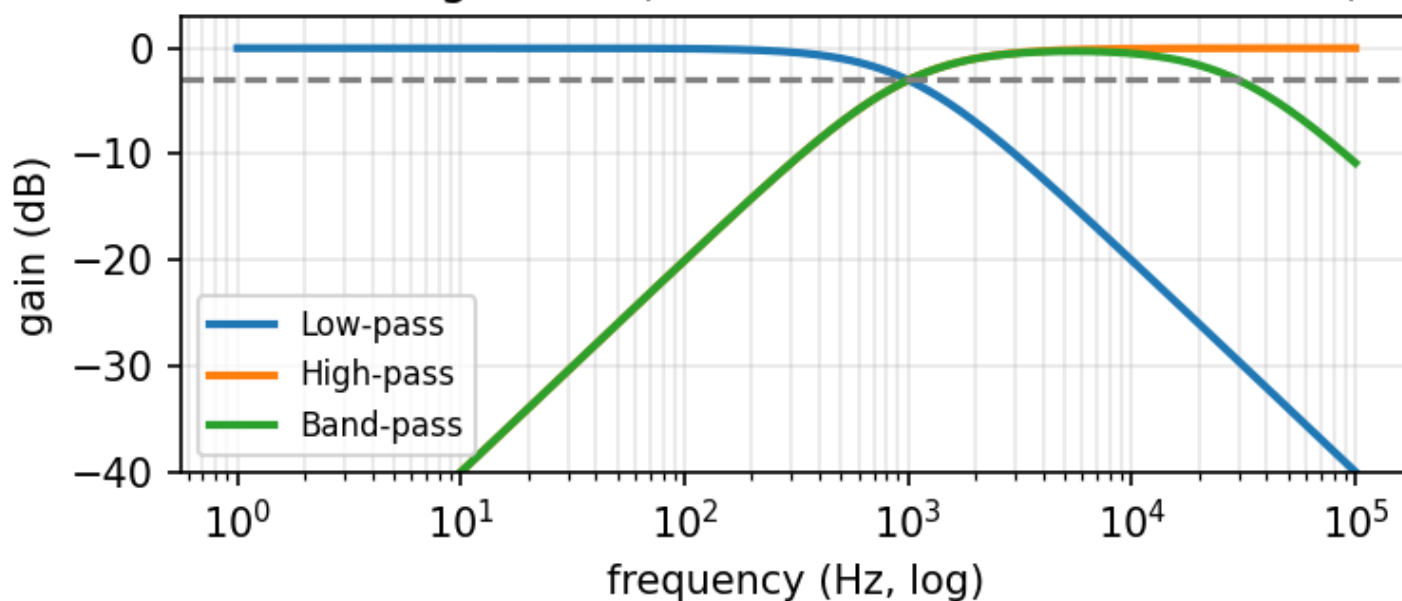
Real $P = V_{\text{rms}} I_{\text{rms}} \cos\theta$ (W) $\leftarrow \cos\theta = \text{power factor (pf)}$
 React. $Q = V_{\text{rms}} I_{\text{rms}} \sin\theta$ (VAR) (+ inductive/lagging, - capacitive/leading)
 Apparent $S = V_{\text{rms}} I_{\text{rms}}$ (VA) Complex $S = V \cdot I^* = P + jQ$; $|S| = \sqrt{P^2 + Q^2}$
 Per element: $P = I_{\text{rms}}^2 R$, $Q = I_{\text{rms}}^2 X$
 PF correction (add parallel C): $Q_C = P(\tan\theta_{\text{old}} - \tan\theta_{\text{new}})$, $C = Q_C / (\omega V_{\text{rms}}^2)$

TYPE 12 — RESONANCE & FILTERS

Series resonance (higher Q = sharper)



Filter magnitude (cutoff at -3 dB, $f_c = 1/2\pi RC$)



Resonance (series RLC): $\omega_0 = 1/\sqrt{LC}$; at ω_0 , Z is purely real ($=R$), current is MAX.
 Quality factor: $Q = \omega_0 L/R = 1/(\omega_0 RC)$ Bandwidth: $BW = \omega_0/Q$
 RC filter cutoff: $f_c = 1/(2\pi RC)$ (gain is 0.707 / -3 dB at f_c)
 Low-pass: output across C. High-pass: output across R. Band-pass: cascade LP+HP.
 Band-pass center (geometric mean): $f_r = \sqrt{f_L \cdot f_H}$

TYPE 13 — NODAL / MESH (when there's no single divider trick)

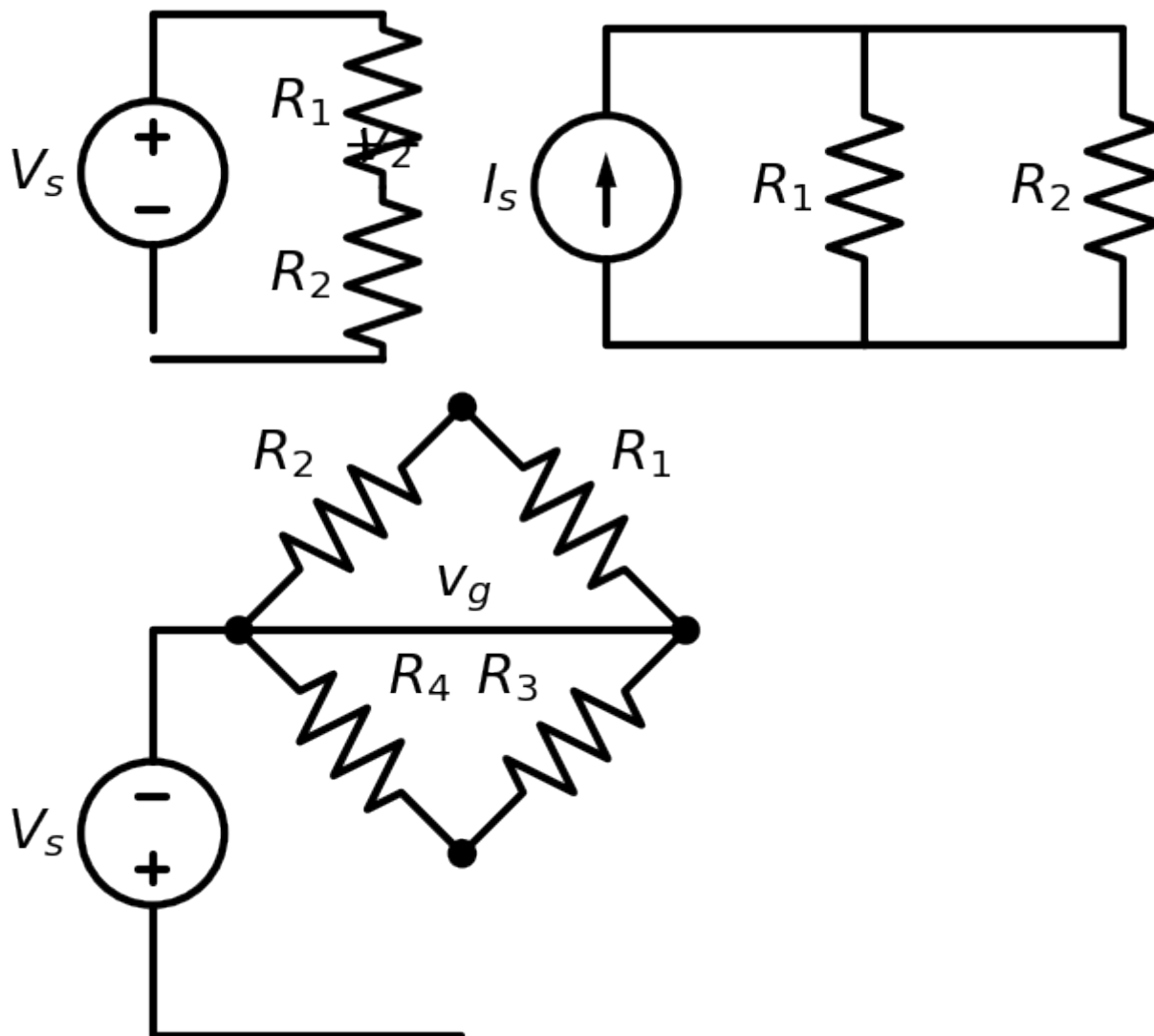
NODAL (pick when fewer nodes):

1. Ground the bottom node.
2. Node voltages $V_1, V_2, \dots$
3. KCL each node: $\sum (V_{\text{this}} - V_{\text{other}})/R = \text{sources}$
4. Supernode if a V-source links two nodes
5. Solve the linear system

MESH (pick when fewer loops):

1. Clockwise mesh currents $i_1, i_2, \dots$
2. KVL each mesh; shared R carries $(i_{\text{this}} - i_{\text{adj}})$
3. Supermesh if a current source is shared
4. Solve the linear system

TYPE 14 — DIVIDERS, Δ -Y, & WHEATSTONE BRIDGE



Voltage divider (series): $v_k = V_s \cdot R_k / \Sigma R$

Current divider (2 parallel): $i_1 = i_{\text{total}} \cdot R_2 / (R_1 + R_2)$ (current favors SMALLER R)

$\Delta \rightarrow Y$: $R_Y = (\text{product of the two adjacent } \Delta \text{ resistors}) / (\text{sum of all three } \Delta \text{ resistors})$

$Y \rightarrow \Delta$: $R_\Delta = (\text{sum of pairwise products of Y resistors}) / (\text{the opposite Y resistor})$

Balanced Wheatstone bridge ($v_g = 0$): $R_1/R_2 = R_3/R_4$ (use to find an unknown R)

TYPE 15 — CHARGE & ENERGY FROM A GRAPH

Charge transferred: $q = \int i \, dt$ = AREA under the i - t graph

Energy: $W = \int p \, dt = \int v \cdot i \, dt$

Method: break the graph into straight-line pieces; write $v(t), i(t)$ for each interval;
multiply $p=v \cdot i$; integrate each interval; add them up.

Diagrams generated for this guide; component values are placeholders — drop in your exam's numbers. Full algebra for each worked example is in *WORKED_EXAMPLES.md*; all formulas in *CHEATSHEET.md*.

ENGR&204 — ULTIMATE FINAL EXAM CHEAT SHEET

Electric Circuits (Green River College) — unlimited pages, everything you need

Tailored to your uploaded files. Textbook = **Dorf & Svoboda, Introduction to Electric Circuits** (your solutions cite P7.5-13, P8.3-25, P8.7-1, etc.). Course flow (from your slides): nodal/mesh → source transform/Thévenin/Norton/superposition/max-power → op-amps → **first-order RC/RL** → **second-order RLC**. Past exams run **4-5 short problems**.

Emphasis is weighted toward the **end of quarter**: first-order **RC/RL** transients and second-order **RLC** circuits. DC tools (nodal/mesh/Thévenin) are included because every transient reduces to a DC problem at $t=0^-$ and $t=\infty$.

Companion file: `WORKED_EXAMPLES.md` has full step-by-step solutions to your actual exam/HW problems (and the RLC handout problems that were left unsolved). Use this file for formulas/method; use that file to pattern-match.

0. EXAM-DAY GAME PLAN (read first)

4-5 short problems. Based on your past finals + Exam 4, expect a mix heavily weighted to the back half: 1. **First-order RC or RL transient** — switch flips → find $v(t)$ or $i(t)$. (Almost guaranteed; see `WORKED_EXAMPLES.md` A1-A6.) 2. **Second-order RLC** — characteristic equation / roots / damping case, or full response. (See B1-B4.) 3. **RC/RL with a twist** — dependent source, op-amp, or reading L/R/C off a $v(t)$ plot. 4. **One earlier-quarter problem** — op-amp, superposition, Thévenin/Norton, power, or equivalent resistance (Part C).

Possible extras some quarters: AC phasors / AC power (§9-10), filters (§9d), charge-energy from a graph.

Universal attack for ANY problem: 1. Label every node and a ground. Mark current directions and $+$ / $-$ on each element (passive sign convention). 2. Identify what type it is (DC steady, transient, or AC). Pick the matching method below. 3. Reduce: combine series/parallel, kill independent sources for Thevenin R, etc. 4. Solve the smallest system you can (one node equation beats three). 5. **Sanity check:** units, signs, limiting behavior ($t=0^+$, $t=\infty$), energy ≥ 0 , power balance ($\sum P_{\text{delivered}} = \sum P_{\text{absorbed}}$).

1. UNITS, CONSTANTS, SIGN CONVENTIONS

Quantity	Symbol	Unit
Voltage	V	volt (V) = J/C
Current	I	ampere (A) = C/s
Resistance	R	ohm (Ω) = V/A
Conductance	$G = 1/R$	siemens (S)
Capacitance	C	farad (F) = C/V
Inductance	L	henry (H) = V·s/A
Power	P	watt (W) = J/s
Energy	W	joule (J)
Charge	q	coulomb (C)
Frequency	f	hertz (Hz)
Ang. freq	ω	rad/s

Metric prefixes: $p=10^{-12}$, $n=10^{-9}$, $\mu=10^{-6}$, $m=10^{-3}$, $k=10^3$, $M=10^6$, $G=10^9$.

Passive sign convention: current enters the $+$ terminal of an element. Then $p = v \cdot i$ is power absorbed ($p > 0$ absorbs, $p < 0$ delivers). Sources usually deliver ($p < 0$).

2. CORE LAWS & ELEMENT EQUATIONS

Ohm's law: $v = i \cdot R$, $i = v/R$.

Power: $P = v \cdot i = i^2 R = v^2/R$.

KCL (nodes): $\sum$ currents leaving a node = 0. **KVL (loops):** $\sum$ voltage drops around a loop = 0.

Element v-i relationships (MEMORIZE)

Element	v-i relation	DC steady state	Stored energy
Resistor	$v = iR$	acts as R	none (dissipates $i^2 R$)
Capacitor	$i = C \, dv/dt$, $v = (1/C) \int i \, dt + v(0)$	open circuit ($i=0$)	$W = \frac{1}{2} C \, v^2$
Inductor	$v = L \, di/dt$, $i = (1/L) \int v \, dt + i(0)$	short circuit ($v=0$)	$W = \frac{1}{2} L \, i^2$

Continuity rules (the keys to transients): - **Capacitor voltage cannot jump:** $v_C(0^+) = v_C(0^-)$. - **Inductor current cannot jump:** $i_L(0^+) = i_L(0^-)$. (Everything else — resistor voltages, source-fed currents — CAN jump instantly.)

3. SERIES / PARALLEL & DIVIDERS

Resistors: series add $R = R_1 + R_2 + \dots$; parallel $1/R = \sum 1/R_k$, two: $R = R_1 R_2 / (R_1 + R_2)$.

Capacitors: opposite of resistors — series $1/C = \sum 1/C_k$; parallel $C = \sum C_k$.

Inductors: like resistors — series add; parallel reciprocal.

Voltage divider (series): $v_k = v_{\text{total}} \cdot R_k / \sum R$.

Current divider (two parallel R): $i_1 = i_{\text{total}} \cdot R_2 / (R_1 + R_2)$ (current favors the smaller R).

4. SYSTEMATIC DC ANALYSIS

4a. Nodal analysis (best when more voltage unknowns)

1. Pick a ground (reference) node.
2. Label node voltages $v_1, v_2, \dots$
3. Write KCL at each non-reference node: $\sum$ currents leaving = 0, using $(v_{\text{this}} - v_{\text{other}})/R$.
4. **Supernode:** if a voltage source sits between two non-ref nodes, enclose both; write one KCL for the pair + the constraint $v_a - v_b = V_{\text{source}}$.
5. Solve the linear system.

4b. Mesh analysis (best for planar, more current unknowns)

1. Define clockwise mesh currents $i_1, i_2, \dots$
2. KVL around each mesh; shared resistor carries $(i_{\text{this}} - i_{\text{adjacent}})$.
3. **Supermesh:** if a current source is shared between two meshes, write KVL around the outer loop + constraint from the current source.

4c. Source transformation

Voltage source V_s in series with $R \rightleftharpoons$ current source $I_s = V_s/R$ in parallel with same R . Use to simplify before nodal/mesh.

4d. Superposition (linear circuits)

Response = sum of responses with **one independent source active at a time**. Deactivate others: **voltage source** $\rightarrow$ **short**, **current source** $\rightarrow$ **open**. (Leave dependent sources ON.)

5. THÉVENIN / NORTON + MAX POWER

Goal: replace everything seen from terminals a-b by V_{th} in series with R_{th} (Thévenin) or I_N in parallel with R_{th} (Norton). $I_N = V_{\text{th}}/R_{\text{th}}$.

Find V_{th} : open-circuit voltage across a-b ($V_{\text{th}} = V_{\text{oc}}$). **Find I_N :** short-circuit current a-b ($I_N = I_{\text{sc}}$). **Find R_{th} :** - No dependent sources: deactivate all independent sources ($V \rightarrow$ short, $I \rightarrow$ open) and compute equivalent R at a-b. - With dependent sources: $R_{\text{th}} = V_{\text{oc}} / I_{\text{sc}}$, OR apply a 1 A test source at a-b with independents off and $R_{\text{th}} = v_{\text{test}} / 1$.

Maximum power transfer: load gets max power when $R_L = R_{\text{th}}$. Then $P_{\text{max}} = V_{\text{th}}^2 / (4 \cdot R_{\text{th}})$.

(AC version: $Z_L = Z_{\text{th}}^*$ (complex conjugate); $P_{\text{max}} = |V_{\text{th}}|^2 / (8 \cdot R_{\text{th}})$ using amplitudes, or $/(4 R_{\text{th}})$ using RMS.)

6. OP-AMPS (ideal) — if covered

Ideal rules: (1) no current into inputs ($i_+ = i_- = 0$); (2) with negative feedback, $v_+ = v_-$ (virtual short).

Config	Output
Inverting	$v_o = -(R_f/R_{\text{in}}) \cdot v_{\text{in}}$
Non-inverting	$v_o = (1 + R_f/R_{\text{in}}) \cdot v_{\text{in}}$
Buffer/follower	$v_o = v_{\text{in}}$
Summing (inverting)	$v_o = -R_f(v_1/R_1 + v_2/R_2 + \dots)$
Difference	$v_o = (R_f/R_1)(v_2 - v_1)$ (matched ratios)
Integrator	$v_o = -(1/RC) \int v_{\text{in}} dt$
Differentiator	$v_o = -RC \cdot dv_{\text{in}}/dt$

7. FIRST-ORDER CIRCUITS — RC & RL TRANSIENTS (HIGH YIELD)

A first-order circuit has **one** equivalent C or L. Every such problem is solved by the **same master formula**.

7a. THE MASTER FORMULA (use for EVERY first-order problem)

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{(-t/\tau)} \quad \text{for } t \geq 0$$

where x is any voltage or current, $x(0^+)$ =initial value, $x(\infty)$ =final (steady-state) value, τ =time constant.

7b. Time constant τ

- **RC circuit:** $\tau = R_{th} \cdot C$
- **RL circuit:** $\tau = L / R_{th}$

R_{th} = Thévenin/equivalent resistance seen by the C (or L) **after the switch settles**, with independent sources deactivated.

7c. STEP-BY-STEP RECIPE (works every time)

1. **$t = 0^-$ (before switch):** assume DC steady state. Capacitor = open, inductor = short. Find $v_C(0^-)$ or $i_L(0^-)$.
2. **Apply continuity:** $v_C(0^+) = v_C(0^-)$, $i_L(0^+) = i_L(0^-)$. (These are your initial conditions.)
3. **$t = \infty$ (after switch, settled):** DC again — cap open, ind short. Find $v_C(\infty)$ or $i_L(\infty)$.
4. **τ :** kill independent sources, find R_{th} at the C/L terminals; $\tau = R_{th}C$ or L/R_{th} .
5. **Plug into master formula** for the capacitor voltage / inductor current.
6. **(If asked for another variable)** find that variable's own $x(0^+)$ and $x(\infty)$ and reuse the same τ , OR get it from v_C / i_L via Ohm's/KCL/KVL. Watch out: non-state variables may jump at $t=0^+$, so compute their 0^+ value from the circuit after the switch using the state variable's initial value.

7d. Special named responses

- **Natural (source-free) response:** $x(\infty)=0 \Rightarrow x(t) = x(0) \cdot e^{(-t/\tau)}$ (pure decay).
- **Step response:** general master formula with nonzero $x(\infty)$.

7e. Useful facts & numbers

- After 1τ : 36.8% of initial remains (63.2% of the way to final). After 5τ : $\approx$ fully settled ($<1\%$).
- Slope at start: $dx/dt|_0 = (x(\infty) - x(0))/\tau$.
- **Capacitor current / inductor voltage** during transient: $i_C(t) = C \cdot dv_C/dt$, $v_L(t) = L \cdot di_L/dt$ (these decay as $e^{(-t/\tau)}$ too).

7f. Worked template — RC step (switch closes at $t=0$)

Given source V_s through R to capacitor C (initially uncharged): - $v_C(0^+) = 0$, $v_C(\infty) = V_s$, $\tau = RC$. - $v_C(t) = V_s(1 - e^{(-t/RC)})$ ("charging"). - $i(t) = (V_s/R) \cdot e^{(-t/RC)}$ (jumps to V_s/R at 0^+ , decays to 0).

Discharging (cap charged to V_0 , source removed): $v_C(t) = V_0 \cdot e^{(-t/RC)}$.

7g. Worked template — RL step (switch at $t=0$)

Source V_s , series R , inductor L : - $i_L(0^+) = i_L(0^-)$, $i_L(\infty) = V_s/R$, $\tau = L/R$. -

$i_L(t) = V_s/R + [i_L(0^+) - V_s/R]e^{(-tR/L)}$. - $v_L(t) = L \cdot di_L/dt$ (jumps at 0^+ , decays to 0).

7h. Singularity / switching functions (on your slides)

- **Unit step** $u(t-t_0) = 0$ for $t < t_0$. A source $V_0 \cdot u(t-t_0)$ turns on at t_0 . $5V[1-u(t)]$ = on for $t < 0$, off after.
- **Unit impulse** $\delta(t-t_0)$ = derivative of the step; $\int \delta(t-t_0)dt = 1$, and $\int f(t)\delta(t-t_0)dt = f(t_0)$.
- **Unit ramp** $r(t) = \int u(t)dt = t \cdot u(t)$.
- These just tell you when sources switch; solve each interval as its own first-/second-order problem and match values at the switching instant (continuity of v_C , i_L).

8. SECOND-ORDER CIRCUITS — RLC (HIGHEST YIELD)

Has **both** L and C. The natural response is governed by a quadratic characteristic equation $\rightarrow$ three damping cases.

8a. Standard form & parameters

The describing ODE reduces to:

$$d^2x/dt^2 + 2\alpha \cdot (dx/dt) + \omega_0^2 \cdot x = \text{forcing}$$

- ω_0 = **undamped natural frequency** (resonant frequency): $\omega_0 = 1/\sqrt{LC}$ (both series & parallel).
- α = **neper / damping coefficient**:
- **Series RLC:** $\alpha = R / (2L)$
- **Parallel RLC:** $\alpha = 1 / (2RC)$

- **Damping ratio:** $\zeta = \alpha/\omega_0$.

8b. Characteristic roots

$$s = -\alpha \pm \sqrt{(\alpha^2 - \omega_0^2)}$$

8c. THE THREE CASES (compare α vs ω_0)

Case	Condition	Roots	Natural response form
Overdamped	$\alpha > \omega_0$	real distinct s_1, s_2	$x = A_1 e^{s_1 t} + A_2 e^{s_2 t}$
Critically damped	$\alpha = \omega_0$	real repeated $s = -\alpha$	$x = (A_1 + A_2 t) e^{-\alpha t}$
Underdamped	$\alpha < \omega_0$	complex $-\alpha \pm j\omega_d$	$x = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$

Damped natural frequency: $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$ (underdamped only).

8d. Complete (total) response

$x(t) = x_{\text{forced}}(\infty) + x_{\text{natural}}(t)$ — add the steady-state (forced) value to the natural form, then solve for the constants using **two** initial conditions.

8e. STEP-BY-STEP RECIPE (RLC)

1. **t=0⁻:** DC steady state → get $v_C(0^-)$ and $i_L(0^-)$ (cap open, ind short).
2. **Initial conditions at 0⁺:** $v_C(0^+) = v_C(0^-)$, $i_L(0^+) = i_L(0^-)$. Get the **derivative** initial condition from the element laws: $-dv_C/dt|_{0^+} = i_C(0^+)/C$ (find $i_C(0^+)$ by KCL at 0⁺). $-di_L/dt|_{0^+} = v_L(0^+)/L$ (find $v_L(0^+)$ by KVL at 0⁺).
3. **Decide series vs parallel** → compute α and ω_0 .
4. **Compare α , ω_0** → pick the case & write the natural-response form.
5. **Final value** $x(\infty)$: DC steady state after switch.
6. **Apply both ICs** [$x(0^+)$ and $x'(0^+)$] to solve for the two constants.
7. Write $x(t)$. Sanity check limits and continuity.

8f. Series vs Parallel summary — INSTRUCTOR'S EXACT TABLE

(Reproduced from your "Review Summary" handout — memorize this.)

Quantity	Parallel RLC	Series RLC
Differential eq.	$d^2i/dt^2 + (1/RC) \cdot di/dt + (1/LC) \cdot i = 0$	$d^2v/dt^2 + (R/L) \cdot dv/dt + (1/LC) \cdot v = 0$
Characteristic eq.	$s^2 + (1/RC)s + 1/LC = 0$	$s^2 + (R/L)s + 1/LC = 0$
Damping coeff. α (rad/s)	$1/(2RC)$	$R/(2L)$
Resonant freq. ω_0 (rad/s)	$1/\sqrt{LC}$	$1/\sqrt{LC}$
Damped freq. ω_d	$\sqrt{(\omega_0^2 - \alpha^2)}$	$\sqrt{(\omega_0^2 - \alpha^2)}$
Overdamped when	$R < \frac{1}{2}\sqrt{L/C}$	$R > 2\sqrt{L/C}$
Critically damped when	$R = \frac{1}{2}\sqrt{L/C}$	$R = 2\sqrt{L/C}$
Underdamped when	$R > \frac{1}{2}\sqrt{L/C}$	$R < 2\sqrt{L/C}$
State var usually solved	node voltage $v(t)$	loop current $i(t)$

Memory hook: "Series has L under α ($R/2L$); Parallel has C under α ($1/2RC$)." The damping condition flips between series and parallel — in **series** a big R is overdamped; in **parallel** a small R is overdamped.

Trick for "find the characteristic equation" problems: kill the independent sources first — the leftover R, L, C usually collapse into a clean series OR parallel loop, then just read α and ω_0 off this table. (Worked: WORKED_EXAMPLES.md B1 = series, B2 = parallel.)

8g. Source-free examples

- **Series, underdamped:** $i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$.
- Find B_1, B_2 from $i(0)$ and $di/dt(0) = v_L(0)/L = (-R \cdot i(0) - v_C(0))/L$ (from KVL).

9. AC STEADY-STATE & PHASORS

9a. Sinusoids

$v(t) = V_m \cos(\omega t + \phi)$. $\omega = 2\pi f$. **Phasor:** $V = V_m \angle \phi$. (Use cosine reference.)

RMS: $V_{\text{rms}} = V_m/\sqrt{2}$ (for sinusoids). Power calcs use RMS.

9b. Impedance Z (Ohm's law becomes $V = I \cdot Z$)

Element	Impedance Z	Phase
Resistor	R	0° (v,i in phase)
Inductor	$j\omega L$	+90° (v leads i)
Capacitor	$1/(j\omega C) = -j/(\omega C)$	-90° (i leads v)

$Z = R + jX$ (X =reactance). Admittance $Y = 1/Z = G + jB$. Series/parallel and dividers work exactly like resistors but with complex Z.

Memory hook: "ELI the ICE man" — in an inductor (L) voltage E leads current I (ELI); in a capacitor (C) current I leads voltage E (ICE).

9c. Phasor analysis recipe

1. Convert sources to phasors; convert R,L,C to impedances at the given ω .
2. Solve with any DC method (nodal/mesh/Thevenin/dividers) using **complex** arithmetic.
3. Convert the phasor answer back to time domain: $V_m \angle \phi \rightarrow V_m \cos(\omega t + \phi)$.

Complex arithmetic: add/subtract in rectangular ($a+jb$); multiply/divide in polar (magnitudes $\times/\div$, angles $+/-$). $a+jb = r \angle \theta$, $r = \sqrt{a^2+b^2}$, $\theta = \text{atan2}(b,a)$.

9d. Resonance (series RLC)

At $\omega_0 = 1/\sqrt{LC}$: Z is purely real ($=R$), $|Z|$ minimum, current maximum. Quality factor $Q = \omega_0 L/R = 1/(\omega_0 RC)$; bandwidth $BW = \omega_0/Q$.

10. AC POWER (common Q4 topic)

For RMS phasors $V = V_{\text{rms}} \angle \theta_v$, $I = I_{\text{rms}} \angle \theta_i$, let $\theta = \theta_v - \theta_i$:

- **Average (real) power:** $P = V_{\text{rms}} \cdot I_{\text{rms}} \cdot \cos \theta$ [W] ($\cos \theta$ = power factor).
- **Reactive power:** $Q = V_{\text{rms}} \cdot I_{\text{rms}} \cdot \sin \theta$ [VAR].
- **Apparent power:** $S = V_{\text{rms}} \cdot I_{\text{rms}}$ [VA].
- **Complex power:** $S = V \cdot I^* = P + jQ$, $|S| = \sqrt{P^2+Q^2}$.
- **Power factor:** $\text{pf} = \cos \theta = P/S$. Lagging (inductive, $\theta > 0$, current lags) or leading (capacitive, $\theta < 0$).
- Per element: $P = I_{\text{rms}}^2 \cdot R$, $Q = I_{\text{rms}}^2 \cdot X$.
- Resistor only absorbs P; ideal L and C only exchange Q ($P=0$).

Power-factor correction: add a parallel **capacitor** to supply Q and reduce line current. Needed cap: $C = Q_C / (\omega \cdot V_{\text{rms}}^2)$ where $Q_C = P(\tan \theta_{\text{old}} - \tan \theta_{\text{new}})$.

Max average power to load (AC): $Z_L = Z_{\text{th}}^*$; $P_{\text{max}} = |V_{\text{th}}(\text{rms})|^2 / (4R_{\text{th}})$.

10A. DELTA-WYE (Δ -Y / π -T) CONVERSION

For resistor networks that are neither series nor parallel (bridges, ladders). - **$\Delta \rightarrow Y$:** each Y resistor = (product of the two adjacent Δ resistors) / (sum of all three Δ resistors). $R_1 = R_b R_c / (R_a + R_b + R_c)$, etc. - **$Y \rightarrow \Delta$:** each Δ resistor = (sum of pairwise products of Y resistors) / (opposite Y resistor). $R_a = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_1$, etc. - **Balanced:** if all Δ equal R_Δ , then $R_Y = R_\Delta/3$.

10B. COMPLEX NUMBER ARITHMETIC (for phasors)

- Rectangular $a + jb \leftrightarrow$ polar $r \angle \theta$: $r = \sqrt{a^2+b^2}$, $\theta = \text{atan2}(b,a)$; $a = r \cos \theta$, $b = r \sin \theta$.
- **Add/subtract:** use rectangular. **Multiply/divide:** use polar ($r_1 r_2 \angle (\theta_1 + \theta_2)$, $(r_1/r_2) \angle (\theta_1 - \theta_2)$).
- $j = 1 \angle 90^\circ$, $1/j = -j = 1 \angle -90^\circ$, $j^2 = -1$.
- Conjugate of $a+jb$ is $a-jb$ ($r \angle -\theta$). Needed for complex power $S = V \cdot I^*$.
- **Calculator:** set to the right mode (DEG vs RAD) and use rectangular $\leftrightarrow$ polar buttons.

10C. SECOND-ORDER INITIAL CONDITIONS (the two you must find)

To solve for the constants you need $x(0^+)$ and $x'(0^+) = dx/dt|_{0^+}$: - Capacitor: $dv_C/dt|_{0^+} = i_C(0^+)/C$ — find $i_C(0^+)$ by KCL at $t=0^+$ (using $i_L(0^+)$, $v_C(0^+)$). - Inductor: $di_L/dt|_{0^+} = v_L(0^+)/L$ — find $v_L(0^+)$ by KVL at $t=0^+$. - Then for the overdamped/critical/underdamped form, solve the 2×2 system from $x(0^+)$ and $x'(0^+)$. - **Complete response:** $x(t) = x_{\text{forced}}(\infty) + x_{\text{natural}}(t)$; apply ICs to the **whole** expression.

10D. RESONANCE, Q, BANDWIDTH, FILTERS

- Series RLC resonance $\omega_0 = 1/\sqrt{LC}$: at ω_0 the reactances cancel, $Z = R$ (min), current max.
- Quality factor: series $Q = \omega_0 L/R = 1/(\omega_0 RC)$; bandwidth $BW = \omega_0/Q = R/L$ (series).
- RC/RL first-order filter cutoff: $f_c = 1/(2\pi RC)$ or $R/(2\pi L)$; gain = 0.707 (-3 dB) at f_c .

- Low-pass = output across C (or R for RL); high-pass = output across R (or L); band-pass = LP cascaded with HP, center $f_r = \sqrt{f_L f_H}$.

10E. CHARGE & ENERGY (graph problems)

- $q = \int i \, dt$ = area under the current-time curve.
- $W = \int p \, dt = \int v \cdot i \, dt$. Split a piecewise-linear graph into intervals, form $p = v \cdot i$ on each, integrate, sum.
- Power sign: $p > 0$ absorbed (into the element), $p < 0$ delivered.

11. QUICK FORMULA WALL (grab-and-go)

Ohm: $V=IR$ $P=VI=I^2R=V^2/R$
 Cap: $i=C \, dv/dt$ $W=\frac{1}{2}CV^2$ (DC: open)
 Ind: $v=L \, di/dt$ $W=\frac{1}{2}LI^2$ (DC: short)
 Continuity: $v_C(0^+)=v_C(0^-)$ $i_L(0^+)=i_L(0^-)$
 1st-order: $x(t)=x(\infty)+[x(0^+)-x(\infty)]e^{-(t/\tau)}$
 $\tau_{RC}=R_{th}C$ $\tau_{RL}=L/R_{th}$
 2nd-order: $\omega_0=1/\sqrt{LC}$
 $\alpha_{series}=R/2L$ $\alpha_{parallel}=1/(2RC)$
 $s=-\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$ $\omega_d=\sqrt{\omega_0^2 - \alpha^2}$
 over: $\alpha > \omega_0$ crit: $\alpha = \omega_0$ under: $\alpha < \omega_0$
 Impedance: $Z_R=R$ $Z_L=j\omega L$ $Z_C=1/(j\omega C)=-j/(\omega C)$
 Phasor power: $P=V_{rms} I_{rms} \cos\theta$ $Q=V_{rms} I_{rms} \sin\theta$ $S=VI^*=P+jQ$
 $pf=\cos\theta=P/S$
 Max power (DC): $R_L=R_{th}$ $P_{max}=V_{th}^2/(4R_{th})$
 Thevenin: $V_{th}=V_{oc}$ $R_{th}=V_{oc}/I_{sc}$ Norton: $I_N=V_{th}/R_{th}$
 RMS sinusoid: $V_{rms}=V_m/\sqrt{2}$
 Resonance: $\omega_0=1/\sqrt{LC}$ $Q=\omega_0 L/R=1/(\omega_0 RC)$ $BW=\omega_0/Q$
 Filter cutoff: $f_c=1/(2\pi RC)$ (-3dB, gain 0.707)
 Delta-Y: $R_Y = (\text{adjacent } \Delta \text{ product})/(\text{sum of all 3 } \Delta)$
 PF correction: $Q_c=P(\tan\theta_{old}-\tan\theta_{new})$ $C=Q_c/(\omega V_{rms}^2)$
 Charge/energy: $q=\int i \, dt$ $W=\int v \cdot i \, dt$
 2nd-order ICs: $dv_C/dt(0)=i_C(0)/C$ $di_L/dt(0)=v_L(0)/L$
 Complex: $a+jb=r\angle\theta$ $r=\sqrt{a^2+b^2}$ $\theta=\text{atan2}(b,a)$ $j=1\angle90^\circ$

12. COMMON MISTAKES / EXAM TRAPS

- Forgetting cap=open / ind=short at DC (both 0^- and ∞). This is half of transient problems.
- Letting v_C or i_L "jump" — they never do. Resistor voltages CAN jump.
- Using the wrong α formula (series vs parallel) — check L vs C in the denominator.
- Mixing peak and RMS in power formulas. Power uses RMS.
- Sign of current direction vs passive sign convention.
- Phasor answers: don't forget to convert back to time domain if asked for $v(t)/i(t)$.
- Radians vs degrees in calculator for phasor trig.
- Time constant uses R_{th} seen by the element, not just the nearest resistor.

13. PREDICTED FINAL & WHERE TO LOOK

Based on your uploaded Exam 4, Homework 6, and the In-Class Prep RLC packet, here's the most likely question lineup and the matching worked example:

Likely question	Method (this file)	Worked example
First-order RC, switch $\rightarrow v(t)$	§7c recipe + master formula	WORKED_EXAMPLES.md A4
First-order RL, find L/R from a plot	§7c + read $x(0^+), x(\infty)$	A1 (Dorf P8.3-25)
First-order w/ dependent source or op-amp	§7c, derive ODE	A2, A3, A5, A6
2nd-order RLC: char. eq. + roots + damping case	§8 + instructor table §8f	B1 (series), B2 (parallel)
2nd-order full / forced response	§8d-8e	B3
RLC design (pick L,C for energy/time spec)	§2 energy + §8	B4 (airbag)
Op-amp / superposition / Thévenin / power	§4-§6	Part C

The 5 things you must not forget on exam day: 1. $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-(t/\tau)}$ — first order, everything. 2. v_C and i_L are continuous (don't jump); everything else can jump. 3. DC steady state: **cap = open**,

ind = short (used at $t=0^-$ and $t=\infty$). 4. RLC: kill sources $\rightarrow$ series or parallel $\rightarrow \alpha, \omega_0 \rightarrow$ compare $\rightarrow$ pick the case. 5. $\alpha_{\text{series}} = R/2L, \alpha_{\text{parallel}} = 1/2RC, \omega_0 = 1/\sqrt{LC}$ for both.

Tailored to your uploaded materials (Dorf/Svoboda textbook, instructor Jae Suk's format). Re-upload anything new and I'll extend this — e.g. more AC phasor problems, mutual inductance, or Laplace if your section covers them.

ENGR&204 — WORKED EXAMPLES (pattern-match your exam to these)

These are fully worked, step-by-step solutions built from **your own uploaded files** (Exam 4 solutions, Homework 6, the In-Class Prep RLC packet). Your textbook is **Dorf & Svoboda, Introduction to Electric Circuits** (problem numbers like P7.5-13, P8.3-25 are Dorf problems), and your past finals are 4-5 questions. Match each exam problem to the closest example below.

Notation used everywhere (same as your HW): $x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{(-t/\tau)}$ for first order. The * next to an "Unsolved on the handout" tag means I solved it from scratch here.

PART A — FIRST-ORDER (RC / RL) — the $t=0^-$ / $t=\infty$ / τ machine

Every first-order problem = find **3 numbers** then plug into the master formula: 1. initial value $x(0^+)$ (use continuity: v_C and i_L don't jump), 2. final value $x(\infty)$ (DC: cap = open, ind = short), 3. time constant τ ($= R_{th} \cdot C$ or L/R_{th} , with sources killed).

A1. RL step response, find L and R from a graph — Dorf P8.3-25 (your Exam 4, Q2)

Circuit: 20 V source — inductor L — in parallel with 40Ω ; a switch opens at $t=0$ putting R_2 in series. The plot shows $v(t)$: 20 V for $t < 0$, jumps to 100 V at $t=0^+$, decays to 20 V. The point (0.14 s, 60 V) is marked. **Find L and R_2 .**

Step 1 — read the three numbers off the plot. - $v(0^-) = 20$ V (steady state before switch). - $v(0^+) = 100$ V (the jump — resistor voltage CAN jump). - $v(\infty) = 20$ V (settles back). - So $v(t) = 20 + 80 e^{(-t/\tau)}$ (since $v(0^+) - v(\infty) = 100 - 20 = 80$).

Step 2 — use the marked point (0.14 s, 60 V) to get τ (really $a = 1/\tau$).

$$60 = 20 + 80 e^{(-a \cdot 0.14)} \\ 40/80 = e^{(-0.14 a)} \rightarrow \ln(0.5) = -0.14 a \rightarrow a = \ln(60-20 \text{ over } 80)/(-0.14) = 5 \text{ (1/s)}$$

So $1/\tau = 5 \Rightarrow \tau = 0.2$ s.

Step 3 — $i(0^+)$ from continuity. Before $t=0$ inductor is a short, so it carried current; at 0^+ the 100 V appears across the 40Ω : $i(0^+) = 100/40 = 2.5$ A (this is $i_L(0^-) = i_L(0^+)$, continuity).

Step 4 — R_2 from the steady-state before switching. For $t < 0$ the inductor is a short, $40 \parallel R_2$ carries the source; the data gives $40 \parallel R_2 = 20/2.5 = 8 \Omega \Rightarrow R_2 = 10 \Omega$.

Step 5 — L from the time constant. $\tau = L/R_{th}$. Here $a = 1/\tau = R_{th}/L = 40/L$ (the 40Ω is the resistance seen by L for $t > 0$). $5 = 40/L \Rightarrow L = 8$ H.

Answer: $v(t) = 20 + 80e^{(-5t)}$ V, $R_2 = 10 \Omega$, $L = 8$ H. **Lesson:** reading $x(0^+)$, $x(\infty)$ off a plot and one extra point gives you τ , then any unknown component.

A2. RL with a current source + mesh equation — Dorf P7.5-13 (your Exam 4, Q1)

Circuit: 10 A source $\parallel 24 \Omega$, a 24Ω series (with $+v(t)-$), center mesh current i_a , another 24Ω , and a 4 H inductor branch carrying $i(t)$.

Step — write ONE mesh (KVL) equation for the center mesh:

$$24 i_a + 24(i_a - i(t)) + 24(i_a - 10) = 0 \rightarrow i_a = (i(t) + 10)/3$$

With the inductor's transient giving $i(t) = 5 - e^{(-4t)}$ A ($t > 0$), substitute:

$$i_a = 5 - e^{(-4t)} \text{ (after simplification)} \\ v(t) = 24 i_a = 120 - 24 e^{(-4t)} \text{ V for } t > 0$$

Lesson: even transient problems often reduce to a single mesh/node equation; solve the algebra, then attach the $e^{(-t/\tau)}$ decay.

A3. First-order RC with a DEPENDENT source — Dorf P8.7-1 (your Exam 4, Q3)

Circuit ($t < 0$, steady state): 12Ω , a dependent current source $2i_x$, a 3Ω , and 38.5 V source. **Find $v_C(t)$.**

Step 1 — initial condition at 0^- (cap = open). KVL:

$$12 i_x + 3(3 i_x) + 38.5 = 0 \Rightarrow i_x = -1.83 \text{ A}$$

$$v_C(0^-) = -12 i_x = 22 \text{ V} = v_C(0^+) \quad (\text{continuity})$$

Step 2 — for $t > 0$ a $1/36 \text{ F}$ cap is switched in and a forcing term $v_1(t) = 8e^{(-5t)}u(t)$ appears. Write **KVL + KCL** with the dependent source:

$$\text{KVL: } 12 i_x(t) - 8e^{(-5t)} + v_C(t) = 0$$

$$\text{KCL: } -i_x(t) - 2i_x(t) + (1/36) dv_C/dt = 0 \Rightarrow i_x(t) = (1/108) dv_C/dt$$

Combine $\rightarrow$ first-order ODE:

$$dv_C/dt + 9 v_C = 72 e^{(-5t)}$$

Step 3 — natural + forced. - Natural: $v_{cn} = A e^{(-9t)}$ (root $s = -9$). - Forced: try $v_{cf} = B e^{(-5t)}$; substitute $\rightarrow -5B + 9B = 72 \Rightarrow B = 18$. So $v_{cf} = 18 e^{(-5t)}$. - Total: $v_C(t) = A e^{(-9t)} + 18 e^{(-5t)}$.

Step 4 — apply IC. $v_C(0) = 22 = A + 18 \Rightarrow A = 4$.

Answer: $v_C(t) = 4 e^{(-9t)} + 18 e^{(-5t)} \text{ V}$. **Lesson:** with a dependent source you can't just use $\tau = RC$ blindly — derive the ODE $dv/dt + a \cdot v = \text{forcing}$, then natural ($e^{(-at)}$) + forced (match the source form).

A4. Clean RC master-formula drill — your Homework 6, Prob 2

Circuit: 12 V source with three 6Ω resistors and a 250 mF cap; a switch changes the network at $t=0$. Find $v(t)$.

- **$t < 0$ (cap open):** $v_C(0^-) = (6/18) \cdot 12 = 4 \text{ V} \Rightarrow v_C(0^+) = 4 \text{ V}$.
- **$t \rightarrow \infty$ (cap open):** $v_C(\infty) = (6/12) \cdot 12 = 6 \text{ V}$.
- **$\tau = R_{th} \cdot C$** with the source killed: $v(t) = 6 + (4 - 6)e^{(-t/\tau)} = 6 - 2 e^{(-t/\tau)} \text{ V}$.

Lesson: this is the purest form of the master formula — most RC exam problems look exactly like this.

A5. RL master-formula with a dependent source — your Homework 6, Prob 3

- **$t < 0$ (ind short):** loop current $i_a = 12/9 = 4/3 \text{ A} = i_L(0^+)$.
- **$t \rightarrow \infty$:** with the dependent source $2i_a$ active, solve $(2i_a - 12)/6 = i_a \rightarrow i_a = -3 \text{ A}$, giving $v = 2(-3) = -6 \text{ V}$, $i_L(\infty) = -6/3 = -2 \text{ A}$.
- **$\tau = L/R_{th}$:** $R_{th} = (1/6 + 1/3)^{-1} = 3? \rightarrow$ here $\tau = 6/3 = 2 \text{ s}$ ($L=6 \text{ H}$ over $R_{th}=3 \Omega$).
- **Plug in:** $i(t) = -2 + (4/3 - (-2))e^{(-t/2)} = -2 + (10/3) e^{(-t/2)} \text{ A}$.

A6. RC / RL with an OP-AMP — your Homework 6, Probs 4 & 5

Op-amp **outputs have 0Ω resistance**, so the output node value is set by the op-amp; treat the cap/inductor with the resistance it actually sees. - **Prob 4 (RC):** $V_o(0^+) = 5 \text{ V}$, $V_o(\infty) = 10 \text{ V}$, $\tau = RC = 20\text{k}\Omega \cdot 4\mu\text{F} = 80\text{ms} \rightarrow V_o(t) = 10 - 5e^{(-12.5t)} \text{ V}$. - **Prob 5 (RL):** $i_L(0^+) = 0.25 \text{ mA}$, $i_L(\infty) = 0.5 \text{ mA}$, $\tau = L/R = 5\text{H}/20\text{k}\Omega = 1/4000 \text{ s} \rightarrow V_o(t) = 5e^{(-4000t)} \text{ V}$.

PART B — SECOND-ORDER (RLC) — characteristic equation $\rightarrow$ damping case

Recipe: kill independent sources $\rightarrow$ identify series vs parallel $\rightarrow$ compute α and $\omega_0 \rightarrow$ compare $\rightarrow$ write the natural-response form $\rightarrow$ add forced response $\rightarrow$ apply the two ICs $x(0^+)$ and $x'(0^+)$.

	Series RLC	Parallel RLC
Char. eq.	$s^2 + (R/L)s + 1/(LC) = 0$	$s^2 + (1/RC)s + 1/(LC) = 0$
α	$R/(2L)$	$1/(2RC)$
ω_0	$1/\sqrt{LC}$	$1/\sqrt{LC}$

Compare: $\alpha > \omega_0$ overdamped, $\alpha = \omega_0$ critical, $\alpha < \omega_0$ underdamped. $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$.

B1. * Series RLC characteristic equation & roots — In-Class Prep Problem 1

Circuit: current source $i_s \parallel 40 \Omega$, with 100 mH inductor in series feeding a $1/3 \text{ mF}$ capacitor. Find the characteristic equation and roots.

Step 1 — kill the independent source. Open the current source. What's left is a single loop: $40 \Omega - 0.1 \text{ H} - (1/3) \times 10^{-3} \text{ F} \rightarrow$ this is a **series RLC**.

Step 2 — parameters.

$$\alpha = R/(2L) = 40 / (2 \cdot 0.1) = 200 \text{ rad/s}$$

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.1 \cdot (1/3) \times 10^{-3}} = 1/\sqrt{3.33 \times 10^{-5}} = 173.2 \text{ rad/s} \quad (\omega_0^2 = 30000)$$

Step 3 — characteristic equation.

$$s^2 + (R/L)s + 1/(LC) = s^2 + 400s + 30000 = 0$$

Step 4 — roots. $s = -200 \pm \sqrt{(200)^2 - 30000} = -200 \pm \sqrt{10000} = -200 \pm 100.$

$$s_1 = -100, \quad s_2 = -300 \quad (\text{rad/s})$$

Step 5 — case. $\alpha (200) > \omega_0 (173.2) \Rightarrow$ **overdamped**. Natural response form: $x(t) = A_1 e^{(-100t)} + A_2 e^{(-300t)}.$
Matches the handout answer $s^2 + 400s + 3 \times 10^4 = 0, s = -300, -100.$

B2. * Parallel RLC natural frequencies (Volkswagen stop-start) — In-Class Prep Problem 3

Given (after killing the step sources): $R = 10 \, \Omega, L = 1/2 \text{ H}, C = 5 \text{ mF}$ arranged as a **parallel RLC**. Find the characteristic equation and natural frequencies.

Step 1 — parallel parameters.

$$\alpha = 1/(2RC) = 1/(2 \cdot 10 \cdot 0.005) = 10 \text{ rad/s}$$

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.5 \cdot 0.005} = 1/\sqrt{0.0025} = 20 \text{ rad/s} \quad (\omega_0^2 = 400)$$

Step 2 — characteristic equation. $s^2 + (1/RC)s + 1/(LC) = s^2 + 20s + 400 = 0.$

Step 3 — roots. $s = -10 \pm \sqrt{(10)^2 - 400} = -10 \pm \sqrt{(-300)} = -10 \pm j17.3.$ **Step 4 — case.** $\alpha (10) < \omega_0 (20) \Rightarrow$ **underdamped**, damped freq $\omega_d = \sqrt{400 - 100} = 17.3 \text{ rad/s}.$ Natural response: $x(t) = e^{(-10t)}(B_1 \cos 17.3t + B_2 \sin 17.3t).$ Matches the handout answer $s^2 + 20s + 400 = 0, s = -10 \pm j17.3.$

B3. * Second-order FORCED response — In-Class Prep Problem 2

Given: series RLC with $R = 3 \, \Omega, L = 1 \text{ H}, C = 1/2 \text{ F},$ driven by source $i_s = 2e^{(-3t)} \text{ A}.$ Find $i(t)$ for $t > 0.$

Step 1 — natural response. Series: $\alpha = R/2L = 1.5, \omega_0 = 1/\sqrt{LC} = \sqrt{2} = 1.414.$ $\alpha > \omega_0 \Rightarrow$ overdamped. Roots: $s = -1.5 \pm \sqrt{(2.25 - 2)} = -1.5 \pm 0.5 = -1, -2.$ So $i_n = A_1 e^{(-t)} + A_2 e^{(-2t)}.$

Step 2 — forced response. Source $\sim e^{(-3t)};$ try $i_f = K e^{(-3t)}.$ Substituting into the circuit's 2nd-order ODE gives $K = 2,$ so $i_f = 2e^{(-3t)}.$

Step 3 — total + initial conditions. $i(t) = A_1 e^{(-t)} + A_2 e^{(-2t)} + 2e^{(-3t)}.$ Use the two ICs from the pre-switch state — here $i(0)=0$ and $di/dt(0)=10:$

$$i(0) = A_1 + A_2 + 2 = 0$$

$$i'(0) = -A_1 - 2A_2 - 6 = 10$$

$$\Rightarrow A_1 = 12, \quad A_2 = -14$$

Answer: $i(t) = 12e^{(-t)} - 14e^{(-2t)} + 2e^{(-3t)} \text{ A}.$ Matches handout. **Lesson:** complete response = (natural form from roots) + (forced term matching the source); the two unknown constants come from $x(0^+)$ and $x'(0^+).$

B4. * Second-order DESIGN (airbag) — In-Class Prep Problem 4

Setup: 12 V charges C through position 1 for a long time, then at $t=0$ the switch moves to position 2 forming a **source-free parallel RLC** with $R = 4 \, \Omega$ (the igniter). Requirement: at least **1 J** dissipated in R, triggering within **0.1 s**. **Select L and C.**

Step 1 — initial energy. Long time on pos 1 $\Rightarrow v_C(0) = 12 \text{ V}, i_L(0) = 0.$ Stored energy

$$W = \frac{1}{2} C v_C^2 = \frac{1}{2} C (144) = 72C.$$

Step 2 — energy constraint. Need $\geq 1 \text{ J}$ delivered to R:

$$72C \geq 1 \Rightarrow C \geq 1/72 \approx 0.014 \text{ F. Pick } C = 1/16 \text{ F} = 0.0625 \text{ F} \Rightarrow \text{stored} = \frac{1}{2} (1/16) (144) = 4.5 \text{ J (comfortable margin).}$$

Step 3 — speed (pick L). For fast delivery choose L so the response decays within 0.1 s. With $L = 0.065 \text{ H}: \alpha = 1/(2RC) = 1/(2 \cdot 4 \cdot 0.0625) = 2, \omega_0 = 1/\sqrt{LC} = 1/\sqrt{0.065 \cdot 0.0625} \approx 15.7 \Rightarrow$ underdamped, envelope $e^{(-2t)}$ plus oscillation that dumps the energy into R quickly. **Answer (one valid design):** $C = 1/16 \text{ F}, L = 0.065 \text{ H}.$ Matches handout. **Lesson:** design problems = pick component(s) to satisfy an **energy** condition ($\frac{1}{2} C V^2 / \frac{1}{2} L I^2$) and a **speed** condition (via α, τ).

PART C — EARLIER-QUARTER TYPES (lower weight, still fair game)

These appear on your Exam 1 / Exam 2; expect at most one on the final.

- **Charge & energy from a v-i graph (Exam 1 Q1):** $q = \int i \, dt = \text{area under } i$; $W = \int p \, dt = \int v \cdot i \, dt$. Break the graph into linear pieces, integrate each.
- **Power balance / find currents & voltages (Exam 1 Q2, HW3):** KCL/KVL, then $P = i^2 R$; check $\Sigma P_{\text{gen}} = \Sigma P_{\text{abs}}$.
- **Find R so that v_a = target (Exam 1 Q3):** express v_a via divider/node eqn in terms of R, set equal, solve.
- **Op-amp find v_o, i_o (Exam 2 Q1):** ideal rules $i_+ = i_- = 0$, $v_+ = v_-$; node equation at inverting input.
- **Superposition (Exam 2 Q3):** one source at a time (V→short, I→open), add.
- **Norton/Thévenin general expression (Exam 2 Q5):** $V_{\text{th}} = V_{\text{oc}}$, $R_{\text{th}} = V_{\text{oc}}/I_{\text{sc}}$, then $i = V_{\text{th}}/(R_{\text{th}} + R)$.
- **Equivalent resistance (your HW):** collapse series/parallel from the far end back to the terminals.

PART D — SECOND-ORDER COMPLETE RESPONSE, AC, POWER, Δ-Y (extra coverage)

D1. Source-free underdamped series RLC — complete response with both ICs

Given: series $R = 2 \, \Omega$, $L = 1 \, \text{H}$, $C = 1/4 \, \text{F}$, source removed at $t=0$, with $i(0) = 0$ and $v_C(0) = 10 \, \text{V}$.

Step 1 — parameters. $\alpha = R/2L = 1$, $\omega_0 = 1/\sqrt{LC} = 2$, $\alpha < \omega_0 \Rightarrow$ underdamped, $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{3} = 1.732$.
Form: $i(t) = e^{-(t)}(B_1 \cos 1.732t + B_2 \sin 1.732t)$.

Step 2 — IC #1 (the value). $i(0) = B_1 = 0$.

Step 3 — IC #2 (the derivative, from KVL). Series loop: $L \cdot di/dt + R \cdot i + v_C = 0$, so $di/dt(0^+) = (-R \cdot i(0) - v_C(0))/L = (0 - 10)/1 = -10$. With $B_1=0$: $di/dt(0) = \omega_d \cdot B_2 = 1.732 \cdot B_2 = -10 \Rightarrow B_2 = -5.77$.

Answer: $i(t) = -5.77 e^{-(t)} \sin(1.732 t) \, \text{A}$. (Verified numerically.) **Lesson:** the two constants always come from $x(0^+)$ and $x'(0^+)$; get the derivative IC from the element law + KVL/KCL.

D2. AC steady-state (phasors)

Given: $v(t) = 20 \cos(2000t) \, \text{V}$ across $R = 10 \, \Omega$ in series with $L = 5 \, \text{mH}$. Find $i(t)$.

$$\begin{aligned} Z_L &= j\omega L = j(2000)(0.005) = j10 \, \Omega \\ Z &= R + jZ_L = 10 + j10 = 14.14 \angle 45^\circ \, \Omega \\ I &= V/Z = (20 \angle 0^\circ)/(14.14 \angle 45^\circ) = 1.414 \angle -45^\circ \, \text{A} \\ i(t) &= 1.414 \cos(2000t - 45^\circ) \, \text{A} \quad (\text{current lags} - \text{inductive}) \end{aligned}$$

D3. Power-factor correction

Given: load $P = 1000 \, \text{W}$, $\text{pf} = 0.7$ lagging, $V_{\text{rms}} = 120 \, \text{V}$, $f = 60 \, \text{Hz}$. Correct to $\text{pf} = 0.95$. Find C .

$$\begin{aligned} \theta_{\text{old}} &= \cos^{-1}(0.7) = 45.6^\circ \rightarrow \tan \theta_{\text{old}} = 1.020 \\ \theta_{\text{new}} &= \cos^{-1}(0.95) = 18.2^\circ \rightarrow \tan \theta_{\text{new}} = 0.329 \\ Q_C &= P(\tan \theta_{\text{old}} - \tan \theta_{\text{new}}) = 1000(1.020 - 0.329) = 691 \, \text{VAR} \\ C &= Q_C/(\omega V_{\text{rms}}^2) = 691/(2\pi \cdot 60 \cdot 120^2) = 1.27 \times 10^{-4} \, \text{F} = 127 \, \mu\text{F} \end{aligned}$$

D4. Δ→Y conversion (for bridge / ladder resistance)

Given: a delta with $R_{ab} = 10$, $R_{bc} = 20$, $R_{ca} = 30 \, \Omega$. Convert to Y (sum = 60).

$$\begin{aligned} R_a &= R_{ab} \cdot R_{ca} / 60 = (10 \cdot 30)/60 = 5 \, \Omega \\ R_b &= R_{ab} \cdot R_{bc} / 60 = (10 \cdot 20)/60 = 3.33 \, \Omega \\ R_c &= R_{bc} \cdot R_{ca} / 60 = (20 \cdot 30)/60 = 10 \, \Omega \end{aligned}$$

Now the network is series/parallel → finish normally.

All Part-B "★" and Part-D solutions were worked from scratch and verified numerically. If your exam uses different numbers, follow the same step sequence.